Huawei Access Node HSR-ACC-74 QoS Congestion Event Alarm ALARM_7054 Resolution SOP

Node: HSR-ACC-74 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_7054

Description

The Huawei Access Node HSR-ACC-74 is experiencing a QoS congestion event alarm, indicated by the ALARM_7054 code, which signifies a warning-level alarm. This alarm is typically triggered due to microbursts, interface tail drops, or QoS queue starvation. The underlying protocol mechanics for this issue are rooted in the Huawei Access Node's ability to prioritize and manage traffic flows within the network. The Access Node's QoS functionality is implemented using the Multi-Protocol Label Switching (MPLS) framework, which enables the node to classify, police, and schedule traffic based on predefined policies. However, in the event of excessive congestion or bursty traffic, the QoS queues may become overwhelmed, leading to packet drops and tail drops, ultimately triggering the ALARM_7054 alarm. Furthermore, the Huawei Access Node's hardware architecture, comprising of Line Cards, Centralized Processing Cards, and System Power Supplies, may also contribute to the congestion event. The Line Cards, responsible for forwarding packets, may become overloaded, whereas the Centralized Processing Cards, handling control plane functions, may experience increased processing loads. If these conditions persist, the QoS congestion event alarm will continue to be triggered, requiring prompt resolution to prevent network degradation.

Pre-requisite Checks

- Verify that the HSR-ACC-74 node is powered on and all interfaces are stable.

- Check the redundant link status to ensure that it is operational and not contributing to the congestion event.

- Dispatch remote hands to the HSR Layout Access site to perform additional checks and troubleshooting, if necessary.

Potential Root Causes

Cause	Probability
Microbursts or bursty traffic overwhelming QoS queues	High
Interface tail drops or packet drops due to congestion	Medium
QoS queue starvation due to misconfigured policies or inadequate policing	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos queue statistic, Expected Output: Queue statistics, including queue depth, packet drop count, and average queue length, Description: This command isolates the QoS queue statistics, enabling the identification of congested queues and the root cause of the ALARM_7054 alarm.
```

```
Command: display interface queue statistic, Expected Output: Interface queue statistics, including queue depth, packet drop count, and average queue length, Description: This command isolates the interface queue statistics, enabling the identification of congested queues and the root cause of the ALARM 7054 alarm.

Command: display qos policy, Expected Output: QoS policy configuration, including policing, shaping, and scheduling parameters, Description: This command isolates the QoS policy configuration, enabling the identification of misconfigured policies or inadequate policing, which may be contributing to the congestion event.
```

Resolution Steps

Step 1: Step: Verify and adjust QoS policies, Command: undo qos policy {policy-name}, Description: This step involves verifying the QoS policy configuration and adjusting the policing, shaping, and scheduling parameters to ensure they are properly configured and not contributing to the congestion event.

Step 2: Step: Clear QoS queue statistics, Command: clear qos queue statistic, Description: This step involves clearing the QoS queue statistics to reset the counters and enable accurate monitoring of congestion events.

Step 3: Step: Verify and adjust interface parameters, Command: display interface queue statistic, Description: This step involves verifying the interface queue statistics and adjusting the interface parameters, such as buffer size or packet buffer threshold, to prevent interface tail drops and packet drops.

Post-Check Verification

```
Command: display qos queue statistic, Expected Output: Queue statistics, including queue depth, packet drop count, and average queue length, Description: This command verifies that the QoS queue statistics are accurate and within normal operating ranges.

Command: display interface queue statistic, Expected Output: Interface queue statistics, including queue depth, packet drop count, and average queue length, Description: This command verifies that the interface queue statistics are accurate and within normal operating ranges.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: apply qos policy {original-policy-name}, Description: This step involves restoring the original QoS policy configuration, if necessary, to ensure that the network operates within its previous configuration.

- Step: Clear QoS queue statistics, Command: clear qos queue statistic, Description: This step involves clearing the QoS queue statistics to reset the counters and enable accurate monitoring of congestion events.